Game Design Document

Core Concept

"Robot Controller" is a turn-based visual programming puzzle game where the player leads a pixel-art robot to a goal (in this case a charging station) using code blocks which control the robots movement. Each level presents one robot to lead to a given goal. Each move requires a certain amount of energy, which is limited, challenging the player to find the most energy efficient path for the robot to take.

Main Features

Objective:

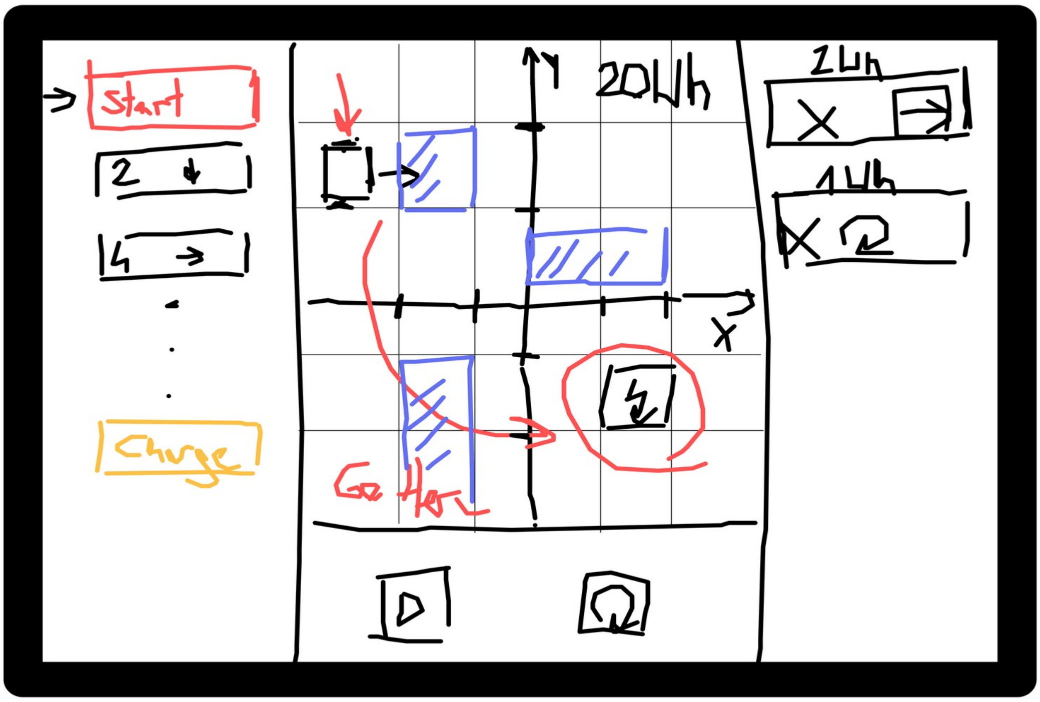
Create a code sequence which leads the robot to the given goal.

- **Goal:** Lead the robot to the charging pad to complete the level
- **Victory Condition:** The robot is on the charging pad, when the sequence is over.

Gameplay Mechanics:

- **Core Interaction:** The player moves various code blocks into a list. When pressing play, these code blocks activate from top to bottom.
- **Player Actions:** The player drags code blocks into the list and then starts the sequence to run the machines. The player is able to stop the sequence if they notice an error and reset the level.
  The player is also able to rearrange the code blocks in the list.
- **Challenges:** The player will face various obstacles, which obstruct the robots movement and have to be navigated around, as well as terrain, which needs more energy to traverse.

- **Sketches:**



- **Story Integration**
  - **Narrative Context:** Players are trapped in a derelict robot repair facility. To escape, they must lead five robots to charging pads, to charge the exit door, so they can escape. The player controls the robots via a computer interface.
  - **Consequences:**
    - Succeed all 5 levels: The exit door unlocks, revealing the next room.
    - Succeed individual level: Player moves to the next level (visual/audio feedback)
    - Failure: The code runs through, but the robot does not reach the goal → The player must restart the level.
- **Pysical or Digital Design**
  - **Set-Up:**
    - Digital screen (monitor or tablet) showing the code block interface and top down view of the room
    - Optional: Physical prop buttons for Play/Restart that trigger the screen
    - Speaker for audio feedback and ambient facility sounds
  - **Visual style:**
    - Retro industrial pixel art aesthetic

        - Color palette: gray for the laboratory area, color coded for code blocks and special terrain.
- **Difficulty and Accessibility**
    - **Skill Level:**

      The game requires players to understand simple programming logic (in what order code is executed) and to think ahead when planning the robots movement. Basic problem-solving skills are necessary to guide the robot through each level. The difficulty increases gradually, allowing players with little or no coding experience to learn the mechanics step by step.

    - **Accessibility Considerations:**

      The game is fully playable using visual information alone. All essential interactions, instructions, and feedback are presented visually, without requiring audio cues. Players do not need sound to understand robot behavior, obstacles, or the results of their code sequences.
- **Technical Details**
    - **Technology Used:**
        - Software: Unity 2D game running on a touchscreen PC or tablet.
        - Display: 16:9 full screen
        - Input: Touch
        - Audio: Built-in speakers or connector sound system
    - **Integration with Escape Room:**
        - The Exit door is offline and needs energy from the robots, which have to stand on a charging pad.

Target Audience

The primary target audience are students in the upper secondary education level, specifically from Gymnasium (Oberstufe) and HTL programs.

The game is designed to introduce programming concepts in an accessible and engaging way, without requiring prior coding knowledge.